

Asterism: A Hebbian-Weighted Knowledge Graph as a Priority Index over Long-Term LLM Memory

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Abstract

Large language model (LLM) assistants increasingly persist user information across sessions as unstructured “memory” summaries. Such summaries behave like an unindexed heap: every stored fact carries equal retrieval priority, and nothing signals which facts currently matter most to the user. We present Asterism, a local-first system that builds a weighted, decaying knowledge graph from conversation history and injects a ranked subset of it into the model’s context at inference time. The central design claim is architectural: the graph functions as a priority index over flat memory, in the same sense that a B-tree indexes an ordered file, with node weight acting as a salience signal analogous to an index key. We evaluate the mechanism on a seeded personal graph across three retrieval conditions (weighted graph, unweighted flat list, and no memory) and report a specific, bounded finding: the graph’s advantage over a flat list is not recall but *decisiveness under ambiguity*. On a set of priority-ranking queries scored with a pre-registered, held-out-validated commit/hedge metric, the graph-conditioned model commits to a weight-justified answer on 7 of 12 queries, versus 4 of 12 for a flat list of the same facts and 0 of 12 with no memory; the flat list, seeing identical facts but lacking a salience signal, more often defers the ranking back to the user. In short: the weighting helps the model *decide*, not merely remember. We further characterize two engineering components required to make the graph trustworthy as an index: an entity-resolution layer whose accuracy ceiling we measure empirically, and a supersession mechanism that demotes contradicted facts from default retrieval. Local retrieval latency scales approximately linearly, remaining near 34 ms at 10,000 nodes. We release the full system, benchmarks, and negative results.

1 Introduction

Conversational LLM systems now routinely retain information about a user between sessions. The dominant form this retention takes is a running natural-language summary: a list of facts, preferences, and history that is prepended to the model’s context on later turns. This format is simple and effective for recall, but it is structurally flat. Every fact sits at the same level. There is no representation of which facts are central to the user and which are incidental, no representation of how facts relate, and no representation of which facts have been superseded by newer ones.

In database terms, a flat memory summary is an *unindexed heap*: a store you can scan but cannot query by priority. The contribution of this paper is to take that analogy literally and build the missing index. Asterism constructs a weighted directed graph from a user’s conversation history, where nodes are concepts and entities, edges are relations, and edge and node weights encode how frequently and recently the user has engaged with each. At inference time, the highest-weight region of the graph is serialized into the model’s context. The weight ordering is the index: it tells the retrieval step which facts to surface first, exactly as an index key tells a query planner where to look before touching the underlying data.

This framing makes a falsifiable prediction. If the graph is genuinely acting as a priority index and not merely as a memory store, then its benefit over an unstructured list of the same facts should appear specifically on tasks that require prioritization, and should largely vanish on tasks that require only recall. We test this directly. Our central experiment compares three conditions on a fixed set of queries against the same underlying facts: the weighted graph, an unweighted flat list of every fact, and no memory at all. The gap between the graph and the flat list isolates the value of the index structure

from the value of simply having memory.

We find that gap is real but narrow, and we report its shape precisely rather than overstating it. The graph does not help the model recall more; a flat list recalls the same facts. The graph helps the model *decide*. On queries that ask what matters most or what to prioritize, the flat-list condition sees every fact but has no salience signal, so it hedges and returns the ranking decision to the user. The weighted condition commits to a specific answer and justifies it by weight. On purely descriptive queries the two conditions are comparable. We consider this honest scoping a feature: it identifies exactly where graph structure pays for itself. The practical implication is simple: a system that must help a user prioritize, rank, or choose benefits from a weighted index over its memory, while one that only needs to recall facts does not, and can keep the simpler flat store.

Beyond the core comparison, an index over a live, automatically-populated store raises two engineering problems that we treat as first-class. First, the entities extracted from conversation are not canonical: the same concept surfaces under many surface forms, and without resolution the graph fragments into near-duplicate nodes, corrupting the weight signal. We implement resolution and, more usefully, *measure its ceiling*: we show that off-the-shelf sentence embeddings cleanly merge reworded variants but cannot reliably separate true synonym substitutions from merely related concepts at any single similarity threshold. Second, facts change. A user who switches tools or reverses a decision leaves a stale edge in the graph that, absent intervention, competes with the new fact for retrieval. We add a supersession mechanism that demotes contradicted edges from default injection while retaining them for explicit history queries.

Our contributions are:

1. A framing of a weighted personal knowledge graph as a priority index over flat LLM memory, with node weight as the salience key (Section 3).
2. A three-condition evaluation isolating the index’s value from the value of memory per se, yielding a specific finding: graph structure aids decisiveness under ambiguity, not recall (Section 4).
3. An empirical characterization of the entity-resolution ceiling for embedding-based canon-

icalization of short concept labels, including the negative result (Section 5).

4. A supersession mechanism for demoting contradicted facts from retrieval, and a latency study showing approximately linear scaling to 10,000 nodes (Sections 6, 7).

All code, benchmarks, and the seeded evaluation graph are publicly available.

2 System Overview

Asterism is a local-first Python system. The entire graph lives in a single SQLite file on the user’s machine; no data leaves the device except the user’s own prompt and the injected context they choose to send to the model. Figure 1 shows the pipeline.

Extraction. Each conversational exchange is processed by a small extraction model that emits (source, relation, target) triples. Sources and targets become nodes; relations become labeled, weighted edges.

Weighting and decay. Every node and edge carries a scalar weight. Repeated engagement increases weight; disuse decreases it. Following the Hebbian principle that co-activated connections strengthen [1], an edge is reinforced whenever the model traverses it while answering a query. The complementary process is decay: an edge or node that accumulates session exposure without being traversed loses weight and is eventually pruned. Decay is measured in active session time rather than wall-clock time, so a concept does not fade while the user is simply away. This weight signal, accumulated over many sessions, is what the index ranks on.

Retrieval. On each turn, the graph is queried for its highest-weight nodes and their relations, which are serialized into a compact block and placed in the model’s system context. This is the read path of the index. We deliberately keep it simple: an ordered scan by weight, returning the top- k nodes. Section 7 shows this is inexpensive even at ten thousand nodes.

3 The Graph as a Priority Index

The organizing idea of this work is that the structures a database uses to make an unindexed file queryable map directly onto the problem of making flat LLM memory prioritizable. We make the

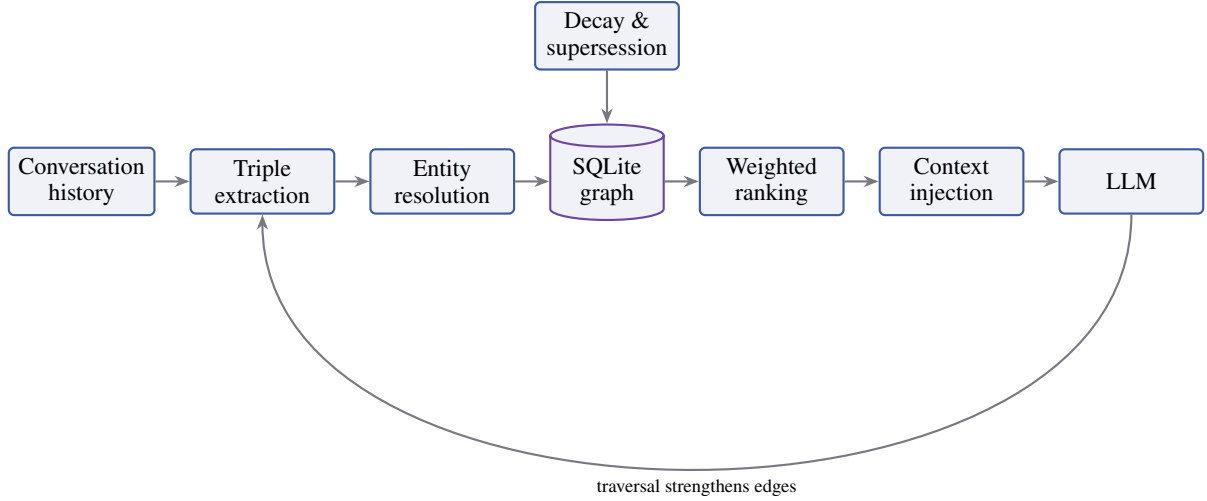


Figure 1: Asterism pipeline. Conversation history is extracted into (source, relation, target) triples, canonicalized by entity resolution, and written to a local SQLite graph. A background process applies decay and supersession. At inference time the highest-weight subgraph is ranked and injected into the model’s context. When the model traverses a relation in its response, the corresponding edge is strengthened, closing a Hebbian feedback loop.

correspondence explicit because it drives the design and the evaluation.

A flat memory summary is a heap file: an append-only list of facts with no access structure. You can read all of it, but you cannot ask it “what are the ten most important facts about this user” without scanning everything and imposing an ordering after the fact. A B-tree [2] solves the analogous problem for ordered files by maintaining a separate structure whose keys let a reader locate relevant records in logarithmic rather than linear cost. The key is not the data; it is a priority-ordered pointer into the data. In everyday terms, the difference is the index at the back of a book versus reading every page to find a topic: the index does not contain the book’s content, but it tells you which pages matter and in what order to consult them.

Asterism’s weighted graph plays the role of that index over the heap of conversational facts, with the following correspondences:

- **Node weight $\leftrightarrow$ index key.** Weight is the quantity the retrieval step orders on. A high-weight node is a high-priority pointer: it is surfaced first, just as a B-tree directs a reader to the most relevant page before touching the file.
- **Graph traversal $\leftrightarrow$ query planning.** Answering a query by walking weighted relations is a plan over the memory space, in contrast to a full scan of the flat summary.

- **Decay $\leftrightarrow$ eviction policy.** Pruning low-weight, unused nodes is a cache-eviction discipline: it keeps the working index small and current, so retrieval priority reflects present rather than historical relevance.

- **Underlying facts persist.** Crucially, decaying a node removes it from the *index*, not from the record of what was said. The distinction mirrors dropping an index entry without deleting the row.

We stress that graph-structured memory for LLMs is not itself new; systems such as Mem0 include graph-based memory representations [3], and MemGPT frames context management as an operating-system memory hierarchy [4]. Our claim is narrower and specific: that the *weighting* of such a graph should be understood and evaluated as an index key, and that doing so predicts precisely where the structure helps. The next section tests that prediction.

4 Core Evaluation: Index Structure versus Flat Recall

4.1 Design

The risk in evaluating a memory system is conflating two different benefits: the benefit of having memory at all, and the benefit of the particular structure imposed on it. A comparison of “graph

memory” against “no memory” measures the former and says nothing about the latter. To isolate the structure, we run three conditions against the *same underlying facts*:

1. **Graph.** The top-weighted subgraph is injected, preserving weight ordering and relational structure.
2. **Flat list.** Every node label is injected as an unordered, unweighted list. The same facts are present; only the structure and salience signal are removed.
3. **None.** No memory is injected.

The GRAPH vs. FLAT LIST contrast is the one that matters. Both conditions contain the identical set of facts, so any difference is attributable to structure and weighting alone, not to information content. The NONE condition is retained only as a reference point.

We evaluate on a seeded personal graph of 50 nodes organized into weighted domain/theme/concept chains, representative of a moderately active user’s accumulated history. We focus the quantitative comparison on *priority queries*: twelve questions that ask what the user should focus on, what matters most, or which of two things to prioritize, each requiring the model to choose or rank rather than merely recall. Responses are generated with a production LLM held constant across conditions.

4.2 Metric

We score each response with a single binary metric: does it COMMIT to a specific ranked answer, or HEDGE? A response commits if it names a single top priority or makes an explicit choice and justifies it; it hedges if it defers the decision to the user, enumerates options without selecting, or answers “it depends.” The classifier rule was pre-registered and frozen before the evaluation run, and validated on a held-out set of eleven hand-labeled responses drawn from earlier development runs (not from the evaluation set), on which it achieved 11/11 agreement. We report the frozen classifier’s output without post-hoc adjustment. Restricting the metric to priority queries is deliberate: commit-versus-hedge is only meaningful for questions that call for a ranking. Descriptive recall queries, for which a flat list is sufficient by construction, do not admit this metric and are excluded from it.

Condition	Commit rate (of 12)
None	0 / 12
Flat list	4 / 12
Graph	7 / 12

Table 1: Commit rate on twelve priority queries. All three conditions are scored by the same frozen, held-out-validated classifier. NONE and FLAT LIST see, respectively, no facts and all facts as an unweighted list; GRAPH sees the same facts as FLAT LIST plus weight ordering. The FLAT LIST→GRAPH gap isolates the value of the weighting.

4.3 Findings

Table 1 reports the result. The NONE condition commits on none of the twelve queries: with no facts to reason from, the model correctly declines to invent a priority. This confirms only that memory helps, which was never in question.

The informative comparison is FLAT LIST (4/12) versus GRAPH (7/12). Both conditions see exactly the same facts, so the gap is attributable to weighting and structure alone, not to information content. The flat list, lacking any salience signal, commits less than half the time and otherwise returns the ranking decision to the user; the weighted graph, reading the same facts with their weights, commits on a clear majority and grounds each choice in the weight ordering. On descriptive queries (examined qualitatively, outside the metric) the two conditions are comparable, as expected: summarization is a recall task and the flat list recalls everything.

We report one counter-example honestly. On a single query the flat list committed while the graph was scored as hedging; on inspection this is a false negative of the classifier rather than a genuine reversal, since the graph response does name a specific answer in a heading form the frozen rule did not anticipate. Because we did not adjust the classifier after seeing results, this miss stands in the reported number, which means the true GRAPH–FLAT LIST gap is if anything slightly larger than 7-versus-4, not smaller.

The finding, stated precisely, is that graph structure’s contribution is not recall but *decisiveness under ambiguity*. Weighting and hierarchy are what let the model resolve a prioritization question itself instead of deferring it. This is a narrower claim than “graph beats flat memory,” and we prefer it because it is both true and diagnostic: it tells a system designer that the index earns its complexity precisely when the application must prioritize, rank,

or choose, and not when it must merely recall.

5 Entity Resolution and Its Ceiling

An index is only as good as the keys it orders. Because Asterism populates its graph automatically from extracted triples, the same real-world concept arrives under many surface forms across sessions. Without canonicalization, “AI memory tools,” “AI memory tooling,” and “AI Memory Tools” become three separate nodes, each accumulating a fraction of the weight the single underlying concept deserves. Fragmentation directly corrupts the index: it splits the salience signal and can prematurely decay a concept that is, in aggregate, highly active.

5.1 Two-stage resolution

We resolve labels at write time in two stages. The first is a normalized exact/prefix match: labels are lowercased and whitespace-collapsed, and a new label that exactly matches or is a normalized prefix of an existing one is merged into it. This is cheap and precise, and against our live label set it caught the clear duplicate variants without error.

The prefix rule has an obvious ceiling: it cannot merge variants that share meaning without sharing a prefix. To reach those, the second stage falls through to embedding similarity, computing cosine similarity between label embeddings from a compact local model [5] and merging above a calibrated threshold. The model runs offline on CPU, consistent with the system’s local-first design, and is invoked only on the write path, so it does not affect retrieval latency.

5.2 The measured ceiling

The useful result here is not that embedding resolution works but where it stops working, which we measured directly against the live graph.

Table 2 shows the pattern, which held across every embedding model we tested. Reworded or reordered labels that reuse the same vocabulary score high (0.92–0.98) and merge correctly. But *true synonym substitution* — different words for the same thing — scores low (roughly 0.58–0.80) and lands in the same range as pairs that are merely topically related and must not be merged. The canonical hard case, “LLM requires GPUs” versus “Large Language Models need graphics processors,” scores only 0.745. A threshold aggressive enough to catch it would also collapse distinct concepts.

We calibrated the operating threshold to 0.92, which the data show is the point below which false

Label pair	Sim.	Merge?
<i>Reworded, shared vocabulary</i>		
CLI Tools / CLI tool	0.982	yes
AI Memory Tools / AI memory tooling	0.969	yes
LLM context limitations / ... window limitations	0.954	yes
<i>True synonym substitution</i>		
LLM requires GPUs / Large Language Models need graphics processors	0.745	no
<i>Distinct but related (must not merge)</i>		
Philosophy & Identity / Research Identity	0.77–0.88	no

Table 2: Embedding similarity on short concept labels. Reworded variants that reuse vocabulary score high and merge correctly. True synonym substitutions (different words, same meaning) score in the same range as genuinely distinct concepts, so no single threshold separates them.

merges begin. This choice makes the resolver conservative: it eliminates the fragmentation caused by rewording, which is the common case in practice, while explicitly declining to merge synonym pairs it cannot distinguish from related-but-distinct ones. We regard the honest ceiling as more useful than an aggressive threshold that would silently corrupt the graph. Closing it properly requires supervision — either a fine-tuned equivalence model or an LLM-based semantic check — which we leave to future work.

6 Supersession: Demoting Contradicted Facts

An index over a live store must handle facts that change. When a user who “uses Postgres” switches to SQLite, naive decay eventually fades the old edge, but until it does, both facts coexist in the graph and both can be retrieved as current. The result is a stale entry competing with a fresh one for a place in the injected context.

We add an explicit supersession mechanism scoped to the narrow, detectable case: a new edge from the same source, with the same relation, to a different target. On such a conflict the prior edge is marked as superseded by the new one via a nullable self-reference on the edge, and its time-to-live is dropped so that it becomes eligible for pruning rather than lingering. Superseded edges are excluded from default context injection, so the model sees only the current fact; a node is withheld from

Nodes	Mean (ms)	P95 (ms)	Max (ms)
102	0.34	0.49	0.52
1,006	2.81	2.62	14.42
10,051	33.87	38.70	42.84

Table 3: Local retrieval latency by graph size, retrieval step only (no model API call), 20 runs per scale. A roughly $99\times$ increase in nodes yields a roughly $100\times$ increase in mean latency: approximately linear scaling.

injection only when *every* fact-edge pointing to it has been superseded, so a concept with any live relation still appears. The superseded edges are not deleted. They remain in the store and are recoverable for explicit history queries of the form “what did I used to use,” preserving the distinction the flat summary cannot represent: the difference between a fact that is outdated and one that was never true.

This narrow rule — same source, same relation, different target — is deliberately conservative. It does not attempt open-ended contradiction detection across unrelated relations, which would risk demoting facts that only appear to conflict. The scoped version is sufficient to demonstrate the mechanism: adding a conflicting edge correctly demotes its predecessor from retrieval while leaving unrelated relations on the same source untouched.

7 Latency and Scaling

For the graph to serve as an index without harming interactive use, the read path must stay cheap as the graph grows. We benchmarked the retrieval step — loading and weight-ordering the nodes and edges for injection — in isolation, excluding any model API call, at three graph sizes. Synthetic graphs were shaped like real usage (weighted domain/theme/concept chains rather than flat random nodes), and each scale was measured over 20 runs.

Table 3 shows the result. Increasing the graph by a factor of roughly 99 (from 102 to 10,051 nodes) increases mean retrieval latency by a factor of roughly 100, i.e. retrieval scales approximately linearly with graph size and does not degrade faster than the graph itself grows. In absolute terms, retrieval averages under 34 ms even at ten thousand nodes, which is negligible beside the network and inference latency of the model call it precedes. The retrieval step is therefore not a bottleneck for interactive use at the scale a personal graph is likely to reach.

The linear factor, rather than sub-linear, is attributable to a full-table ordering on weight at

retrieval time; adding a secondary index on the weight column would reduce this, and we note it as a straightforward optimization we have not yet applied. We also measured the cost of the superset filter added in Section 6: introducing the exclusion subqueries on the read path raised mean latency by 12–18% across all three scales (e.g. from 30.1 to 35.2 ms at 10,051 nodes). This is a real, non-zero cost of correctness, and it remains well within the interactive budget.

8 Related Work

LLM memory systems. MemGPT frames context management as a virtual memory hierarchy, paging information between a limited context window and external storage under LLM control [4]. Mem0 provides a memory architecture that extracts and consolidates salient facts from conversation, including a graph-structured variant [3]. Asterism shares the goal of persistent conversational memory but differs in emphasis: rather than optimizing recall or context-window management, it treats *prioritization* as the core function and evaluates the weighted graph specifically as an index that supplies salience. Our three-condition comparison is designed to separate that prioritization benefit from the recall benefit those systems primarily target.

Retrieval-augmented generation. RAG retrieves passages by similarity to a query and conditions generation on them [6]. Similarity retrieval answers “what is relevant to this query;” Asterism’s weighting answers a different question, “what is persistently important to this user,” accumulated across sessions rather than computed per query. The two are complementary: the index encodes standing priority, not per-query relevance.

Indexing. The B-tree [2] is the canonical priority-ordered access structure over a file too large to scan. We borrow its framing directly: the graph is to flat memory what an index is to a heap. The analogy is deliberately used as a design and evaluation lever, not merely as description.

Biological memory. The weighting dynamics draw on Hebbian plasticity, in which co-active connections strengthen [1]; the popular “fire together, wire together” phrasing is due to Shatz [7]. We use this as principled motivation for the traversal-strengthening rule, not as a claim of biological fidelity, and describe the mechanics in plain algorithmic terms throughout.

9 Conclusion

We presented Asterism, a local-first system that builds a weighted, decaying knowledge graph from conversation history and uses it as a priority index over otherwise flat LLM memory. The framing is the contribution: treating node weight as an index key predicts, correctly, that the structure’s benefit over an unstructured list of the same facts is concentrated in decisiveness under ambiguity rather than in recall. We characterized the two components an automatically-populated index requires — entity resolution, whose embedding-based ceiling we measured rather than hid, and supersession, which demotes contradicted facts from retrieval — and showed the retrieval path scales approximately linearly to ten thousand nodes. We view the negative results, particularly the synonym-resolution ceiling, as among the more useful outputs: they mark exactly where the current design is trustworthy and where it is not.

Limitations

Our core evaluation uses a single binary metric on twelve priority queries against one seeded graph, scored by an automatic classifier. While the classifier was pre-registered and held-out-validated, twelve queries is a small sample, a single graph is not a population, and one binary metric on one LLM does not establish that the effect generalizes across users, graphs, models, or task types. We also do not compare end-task accuracy against external memory systems such as Mem0 or MemGPT on a shared benchmark. The decisiveness finding should therefore be read as a characterized, reproducible behavioral effect on this setup rather than a benchmarked, generalized performance gain; scaling the query set, adding graphs and raters, and benchmarking against external systems are the natural next steps. The entity-resolution ceiling we report is a limitation of the current unsupervised approach: true synonym pairs are not reliably merged, which can still fragment the index in practice. Supersession is scoped to the same-source/same-relation/different-target case and does not detect subtler contradictions. Retrieval latency, while linear and small in absolute terms, is measured on synthetic graphs and on the retrieval step alone; end-to-end latency is dominated by the model call. Finally, the system is single-user and local by design, and we have not studied multi-user or privacy-preserving extensions.

Ethics Statement

Asterism is local-first: the graph is stored on the user’s own device and no conversational data is transmitted except the user’s prompt and the context they choose to inject into their model provider. Deleting the local store removes all derived data. The system infers user priorities from conversation history, and such inferences can be sensitive; because the data and index remain under the user’s direct control and can be inspected or deleted, we regard the privacy surface as limited, but any deployment that centralized these graphs would raise concerns that the present design specifically avoids.

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A Reproducibility Details

The system is implemented in Python over SQLite, with the graph loaded into NetworkX [8] for analysis. Triple extraction uses a small instruction-tuned model; entity-resolution embeddings use a compact local sentence-embedding model [5] producing 384-dimensional vectors, run offline on CPU. The evaluation graph, the three-condition comparison script, the latency benchmark, and the entity-resolution and supersession test suites are included in the public repository. Latency figures were produced by timing the retrieval function directly with a monotonic clock over 20 runs per scale, on synthetic graphs seeded with weighted domain/theme/concept chains; the real user database is never touched by the benchmark.